IMPLEMENTATION OF CNN

Harshit Kumar More

Roll No: 16

Final Accuracy:

BatchSize\Learning rate	0.1	0.01	0.001
16	68.18%	70.69%	58.11%
32	70.94%	68.29%	52.08%
64	67.11%	62.12%	44.07%

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Running with BatchSize=16, LearningRate=0.1
BS=16, LR=0.1, Epoch=1: 100%|██████████| 3125/3125 [01:11<00:00, 43.85it/s, loss=1.13]
Epoch 1: Test Accuracy = 55.09%
BS=16, LR=0.1, Epoch=2: 100%|██████████| 3125/3125 [01:09<00:00, 44.68it/s, loss=1.11]
Epoch 2: Test Accuracy = 60.88%
BS=16, LR=0.1, Epoch=3: 100%|██████████| 3125/3125 [01:09<00:00, 44.65it/s, loss=1.1]
Epoch 3: Test Accuracy = 64.03%
BS=16, LR=0.1, Epoch=4: 100%|██████████| 3125/3125 [01:09<00:00, 44.83it/s, loss=1.34]
Epoch 4: Test Accuracy = 67.44%
BS=16, LR=0.1, Epoch=5: 100%|██████████| 3125/3125 [01:09<00:00, 44.85it/s, loss=1.36]
Epoch 5: Test Accuracy = 68.35%
BS=16, LR=0.1, Epoch=6: 100%|██████████| 3125/3125 [01:10<00:00, 44.37it/s, loss=1.69]
Epoch 6: Test Accuracy = 69.97%
BS=16, LR=0.1, Epoch=7: 100%|██████████| 3125/3125 [01:10<00:00, 44.41it/s, loss=1.16]
Epoch 7: Test Accuracy = 68.18%

Running with BatchSize=16, LearningRate=0.01
BS=16, LR=0.01, Epoch=1: 100%|██████████| 3125/3125 [01:11<00:00, 43.43it/s, loss=1.09]
Epoch 1: Test Accuracy = 56.25%
BS=16, LR=0.01, Epoch=2: 100%|██████████| 3125/3125 [01:12<00:00, 43.15it/s, loss=1.74]
Epoch 2: Test Accuracy = 59.82%
BS=16, LR=0.01, Epoch=3: 100%|██████████| 3125/3125 [01:10<00:00, 44.24it/s, loss=0.967]
Epoch 3: Test Accuracy = 64.56%
BS=16, LR=0.01, Epoch=4: 100%|██████████| 3125/3125 [01:11<00:00, 43.94it/s, loss=0.917]
Epoch 4: Test Accuracy = 65.55%
BS=16, LR=0.01, Epoch=5: 100%|██████████| 3125/3125 [01:11<00:00, 43.44it/s, loss=0.786]
Epoch 5: Test Accuracy = 67.88%
BS=16, LR=0.01, Epoch=6: 100%|██████████| 3125/3125 [01:10<00:00, 44.08it/s, loss=0.947]
Epoch 6: Test Accuracy = 69.00%
BS=16, LR=0.01, Epoch=7: 100%|██████████| 3125/3125 [01:10<00:00, 44.14it/s, loss=1.19]
Epoch 7: Test Accuracy = 70.69%

Running with BatchSize=16, LearningRate=0.001
BS=16, LR=0.001, Epoch=1: 100%|██████████| 3125/3125 [01:10<00:00, 44.16it/s, loss=1.75]
Epoch 1: Test Accuracy = 37.24%
BS=16, LR=0.001, Epoch=2: 100%|██████████| 3125/3125 [01:12<00:00, 42.93it/s, loss=1.56]
Epoch 2: Test Accuracy = 44.80%
BS=16, LR=0.001, Epoch=3: 100%|██████████| 3125/3125 [01:13<00:00, 42.70it/s, loss=1.82]
Epoch 3: Test Accuracy = 49.81%
BS=16, LR=0.001, Epoch=4: 100%|██████████| 3125/3125 [01:12<00:00, 43.34it/s, loss=1.49]
Epoch 4: Test Accuracy = 52.96%
BS=16, LR=0.001, Epoch=5: 100%|██████████| 3125/3125 [01:13<00:00, 42.75it/s, loss=1.4]
Epoch 5: Test Accuracy = 55.74%
BS=16, LR=0.001, Epoch=6: 100%|██████████| 3125/3125 [01:12<00:00, 42.85it/s, loss=1.08]
Epoch 6: Test Accuracy = 57.20%
BS=16, LR=0.001, Epoch=7: 100%|██████████| 3125/3125 [01:12<00:00, 43.07it/s, loss=1.52]
Epoch 7: Test Accuracy = 58.11%
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Running with BatchSize=32, LearningRate=0.1
BS=32, LR=0.1, Epoch=1: 100%|██████████| 1563/1563 [01:00<00:00, 26.03it/s, loss=1.29]
Epoch 1: Test Accuracy = 55.20%
BS=32, LR=0.1, Epoch=2: 100%|██████████| 1563/1563 [00:59<00:00, 26.21it/s, loss=1.68]
Epoch 2: Test Accuracy = 59.32%
BS=32, LR=0.1, Epoch=3: 100%|██████████| 1563/1563 [00:59<00:00, 26.11it/s, loss=1.08]
Epoch 3: Test Accuracy = 62.26%
BS=32, LR=0.1, Epoch=4: 100%|██████████| 1563/1563 [00:59<00:00, 26.44it/s, loss=1.02]
Epoch 4: Test Accuracy = 65.34%
BS=32, LR=0.1, Epoch=5: 100%|██████████| 1563/1563 [00:58<00:00, 26.62it/s, loss=1.23]
Epoch 5: Test Accuracy = 67.60%
BS=32, LR=0.1, Epoch=6: 100%|██████████| 1563/1563 [00:58<00:00, 26.59it/s, loss=1.08]
Epoch 6: Test Accuracy = 71.02%
BS=32, LR=0.1, Epoch=7: 100%|██████████| 1563/1563 [00:59<00:00, 26.17it/s, loss=0.843]
Epoch 7: Test Accuracy = 70.94%

Running with BatchSize=32, LearningRate=0.01
BS=32, LR=0.01, Epoch=1: 100%|██████████| 1563/1563 [00:59<00:00, 26.32it/s, loss=1.42]
Epoch 1: Test Accuracy = 50.70%
BS=32, LR=0.01, Epoch=2: 100%|██████████| 1563/1563 [00:59<00:00, 26.24it/s, loss=1.4]
Epoch 2: Test Accuracy = 55.31%
BS=32, LR=0.01, Epoch=3: 100%|██████████| 1563/1563 [00:59<00:00, 26.18it/s, loss=1.6]
Epoch 3: Test Accuracy = 56.55%
BS=32, LR=0.01, Epoch=4: 100%|██████████| 1563/1563 [00:59<00:00, 26.36it/s, loss=1.06]
Epoch 4: Test Accuracy = 62.62%
BS=32, LR=0.01, Epoch=5: 100%|██████████| 1563/1563 [00:59<00:00, 26.11it/s, loss=1.01]
Epoch 5: Test Accuracy = 65.25%
BS=32, LR=0.01, Epoch=6: 100%|██████████| 1563/1563 [00:59<00:00, 26.07it/s, loss=1.15]
Epoch 6: Test Accuracy = 67.29%
BS=32, LR=0.01, Epoch=7: 100%|██████████| 1563/1563 [00:59<00:00, 26.25it/s, loss=0.969]
Epoch 7: Test Accuracy = 68.29%

Running with BatchSize=32, LearningRate=0.001
BS=32, LR=0.001, Epoch=1: 100%|██████████| 1563/1563 [00:59<00:00, 26.39it/s, loss=1.97]
Epoch 1: Test Accuracy = 34.38%
BS=32, LR=0.001, Epoch=2: 100%|██████████| 1563/1563 [00:59<00:00, 26.09it/s, loss=1.6]
Epoch 2: Test Accuracy = 39.84%
BS=32, LR=0.001, Epoch=3: 100%|██████████| 1563/1563 [00:59<00:00, 26.07it/s, loss=1.73]
Epoch 3: Test Accuracy = 43.27%
BS=32, LR=0.001, Epoch=4: 100%|██████████| 1563/1563 [01:00<00:00, 26.04it/s, loss=1.72]
Epoch 4: Test Accuracy = 46.75%
BS=32, LR=0.001, Epoch=5: 100%|██████████| 1563/1563 [01:00<00:00, 25.74it/s, loss=1.75]
Epoch 5: Test Accuracy = 49.35%
BS=32, LR=0.001, Epoch=6: 100%|██████████| 1563/1563 [01:00<00:00, 25.72it/s, loss=1.32]
Epoch 6: Test Accuracy = 51.06%
BS=32, LR=0.001, Epoch=7: 100%|██████████| 1563/1563 [01:00<00:00, 25.67it/s, loss=1.17]
Epoch 7: Test Accuracy = 52.08%
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Running with BatchSize=64, LearningRate=0.1
BS=64, LR=0.1, Epoch=1: 100%|██████████| 782/782 [00:54<00:00, 14.47it/s, loss=1.38]
Epoch 1: Test Accuracy = 45.20%
BS=64, LR=0.1, Epoch=2: 100%|██████████| 782/782 [00:53<00:00, 14.55it/s, loss=1.23]
Epoch 2: Test Accuracy = 51.28%
BS=64, LR=0.1, Epoch=3: 100%|██████████| 782/782 [00:53<00:00, 14.60it/s, loss=0.973]
Epoch 3: Test Accuracy = 61.86%
BS=64, LR=0.1, Epoch=4: 100%|██████████| 782/782 [00:53<00:00, 14.53it/s, loss=0.867]
Epoch 4: Test Accuracy = 65.75%
BS=64, LR=0.1, Epoch=5: 100%|██████████| 782/782 [00:53<00:00, 14.65it/s, loss=0.725]
Epoch 5: Test Accuracy = 62.69%
BS=64, LR=0.1, Epoch=6: 100%|██████████| 782/782 [00:53<00:00, 14.63it/s, loss=1.17]
Epoch 6: Test Accuracy = 61.87%
BS=64, LR=0.1, Epoch=7: 100%|██████████| 782/782 [00:53<00:00, 14.70it/s, loss=0.97]
Epoch 7: Test Accuracy = 67.11%

Running with BatchSize=64, LearningRate=0.01
BS=64, LR=0.01, Epoch=1: 100%|██████████| 782/782 [00:53<00:00, 14.73it/s, loss=1.54]
Epoch 1: Test Accuracy = 46.42%
BS=64, LR=0.01, Epoch=2: 100%|██████████| 782/782 [00:53<00:00, 14.48it/s, loss=1.81]
Epoch 2: Test Accuracy = 50.13%
BS=64, LR=0.01, Epoch=3: 100%|██████████| 782/782 [00:54<00:00, 14.44it/s, loss=1.63]
Epoch 3: Test Accuracy = 50.84%
BS=64, LR=0.01, Epoch=4: 100%|██████████| 782/782 [00:53<00:00, 14.55it/s, loss=1.06]
Epoch 4: Test Accuracy = 57.65%
BS=64, LR=0.01, Epoch=5: 100%|██████████| 782/782 [00:53<00:00, 14.70it/s, loss=1.61]
Epoch 5: Test Accuracy = 61.23%
BS=64, LR=0.01, Epoch=6: 100%|██████████| 782/782 [00:53<00:00, 14.73it/s, loss=1.52]
Epoch 6: Test Accuracy = 62.17%
BS=64, LR=0.01, Epoch=7: 100%|██████████| 782/782 [00:53<00:00, 14.70it/s, loss=1.14]
Epoch 7: Test Accuracy = 62.12%

Running with BatchSize=64, LearningRate=0.001
BS=64, LR=0.001, Epoch=1: 100%|██████████| 782/782 [00:53<00:00, 14.72it/s, loss=2.22]
Epoch 1: Test Accuracy = 27.05%
BS=64, LR=0.001, Epoch=2: 100%|██████████| 782/782 [00:53<00:00, 14.73it/s, loss=2.06]
Epoch 2: Test Accuracy = 33.65%
BS=64, LR=0.001, Epoch=3: 100%|██████████| 782/782 [00:53<00:00, 14.69it/s, loss=1.77]
Epoch 3: Test Accuracy = 37.75%
BS=64, LR=0.001, Epoch=4: 100%|██████████| 782/782 [00:54<00:00, 14.31it/s, loss=2.15]
Epoch 4: Test Accuracy = 40.01%
BS=64, LR=0.001, Epoch=5: 100%|██████████| 782/782 [00:54<00:00, 14.41it/s, loss=1.65]
Epoch 5: Test Accuracy = 41.52%
BS=64, LR=0.001, Epoch=6: 100%|██████████| 782/782 [00:53<00:00, 14.69it/s, loss=1.36]
Epoch 6: Test Accuracy = 42.75%
BS=64, LR=0.001, Epoch=7: 100%|██████████| 782/782 [00:53<00:00, 14.73it/s, loss=1.55]
Epoch 7: Test Accuracy = 44.07%
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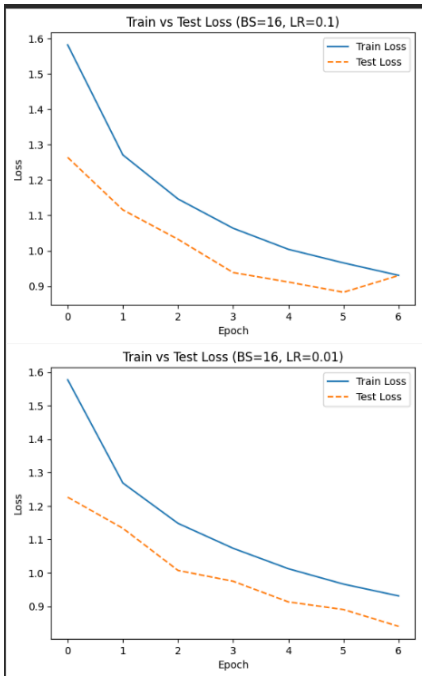
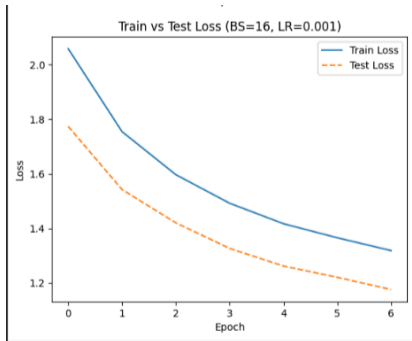
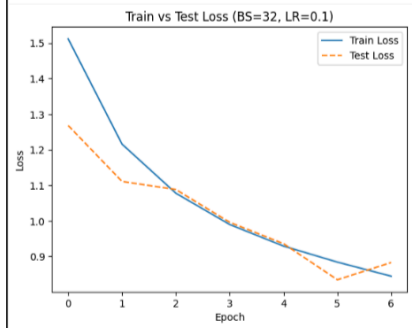
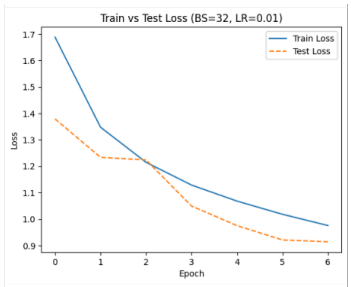
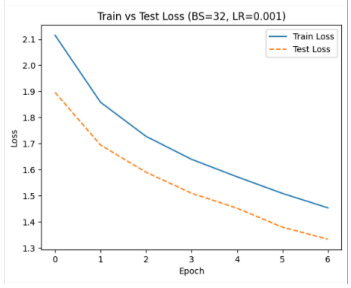

Train vs Test Loss (BS=16, LR=0.1)
Train Loss
Test Loss
Loss
Epoch

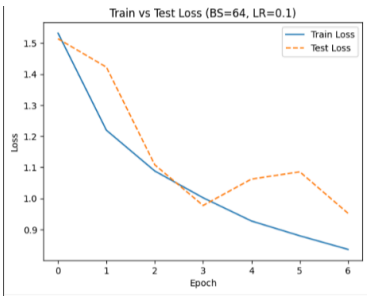

Train vs Test Loss (BS=16, LR=0.001)
Train Loss
Test Loss
Loss
Epoch

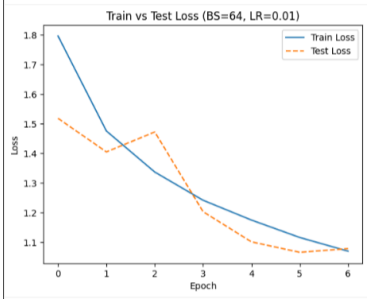
Train vs Test Loss (BS=16, LR=0.01)
Train Loss
Test Loss
Loss
Epoch

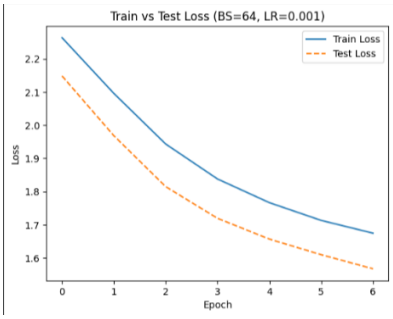

Train vs Test Loss (BS=32, LR=0.1)
Train Loss
Test Loss
Loss
Epoch


Train vs Test Loss (BS=32, LR=0.01)
Train Loss
Test Loss
Loss
Epoch


Train vs Test Loss (BS=32, LR=0.001)
Train Loss
Test Loss
Loss
Epoch


Train vs Test Loss (BS=64, LR=0.1)
Train Loss
Test Loss
Loss
Epoch


Train vs Test Loss (BS=64, LR=0.01)
Train Loss
Test Loss
Loss
Epoch


Train vs Test Loss (BS=64, LR=0.001)
Train Loss
Test Loss
Loss
Epoch